Challenges and opportunities in remote sensing-based fuel load estimation for **wildfire behavior** and **management**: A comprehensive **review**

A Abdollahi, M Yebra - Remote Sensing, 2025 - opus.lib.uts.edu.au

… **review** encompassing various remote-sensing applications for diverse **forest fire** fuel … array of remote-sensing data used for **forest fire** fuels, our **review** delves more deeply into the data …

☆ Save 99 Cite Cited by 10 Related articles All 7 versions »

Advances in **Wildfire Spread** Detection and **Prediction**: Techniques, Challenges, and Applications

RY Zakari, K Kalinaki, ZK Lawal… - Artificial Intelligence and … - taylorfrancis.com

… DL algorithms in revolutionizing **wildfire** detection and **prediction**, enabling … to **forecast wildfire behavior** and **spread** patterns. … Spatial **prediction** of **wildfire** susceptibility using field **survey** …

☆ Save 99 Cite Related articles »

Fast-moving dire **wildfire** evacuation **simulation**

S Grajdura, S Borjigin, D Niemeier - Transportation research part D …, 2022 - Elsevier

… There is a gap in our knowledge of human evacuation **behavior** under these conditions. In … **wildfire** evacuation. We use a post-disaster **survey** and decision tree methods to **model** agent …

☆ Save 99 Cite Cited by 46 Related articles All 5 versions

Modeling **wildland fire propagation** with level set methods

V Mallet, DE Keyes, FE Fendell - Computers & Mathematics with …, 2009 - Elsevier

… burning, and post-crisis **review**, eg, personnel training, … **wildland fire spread** models, especially a semi-empirical, equilibrium-type **model** proposed in [11] for wind-aided fire **spread** …

☆ Save 99 Cite Cited by 142 Related articles All 17 versions

A **Survey** of **Wildfire Spread Prediction** and Risk Estimation Methodswith Machine Learning Techniques

S Dascalu, R Wu, A Munoz, S Guan - 2022 - scholarsarchive.byu.edu

… we conducted a **survey** on papers published recently focused on **wildfire spread prediction** and … In the rest of the paper, Section 2 illustrates how **wildfire spread** behaviour is predicted in …

☆ Save 99 Cite Related articles »

Systematic **review** and research gaps on **wildfire** evacuations: infrastructure, transportation modes, networks, and planning

SN Zehra, SD Wong - Transportation Planning and Technology, 2024 - Taylor & Francis

… a **review** of human-centered topics (eg equity, **management**, decision-making) within the larger **wildfire** … including papers on simulations and **behavior** in **wildfire** evacuations to focus on …

☆ Save 99 Cite Cited by 15 Related articles All 8 versions

A **review** on status, implications and recent trends of **forest fire management**

V Attri, R Dhiman, S Sarvade - Archives of Agriculture and …, 2020 - academia.edu

… **Forest fire spread** out in an area having combustible material … **Forest fire** starts from the ignition point and **spread** outwards … topography were made favorable environment for **forest fire**. …

☆ Save 99 Cite Cited by 65 Related articles All 6 versions »

A data-driven **model** for large **wildfire** behaviour **prediction** in Europe

D Rodriguez-Aseretto, D de Rigo, M Di Leo… - Procedia Computer …, 2013 - Elsevier

… as a required input for a realistic **prediction** of the **wildfire dynamics**. In this case, the fire controls are decided elsewhere and notified to the predictive **model**. This may be exploited by …

☆ Save 99 Cite Cited by 48 Related articles All 12 versions

A **review** of recent forest and **wildland fire management** decision support systems research

DL Martell - Current Forestry Reports, 2015 - Springer

… Fire behaviour or the physics and chemistry of fire ignition and **spread**. … focus of this **review** is very decidedly North America for the period 2010 through 2014. **Wildland fire** and the …

☆ Save 99 Cite Cited by 165 Related articles All 5 versions

Forest fire spread simulation and fire extinguishing visualization research

Q Meng, H Lu, Y Huai, H Xu, S Yang - Forests, 2023 - mdpi.com

… **forest fire spread** quickly. Therefore, this paper proposes a lightweight **forest fire spread model** … ensuring the realism of **forest fire behavior** and the immersion of the virtual environment. …

☆ Save 99 Cite Cited by 19 Related articles All 5 versions »